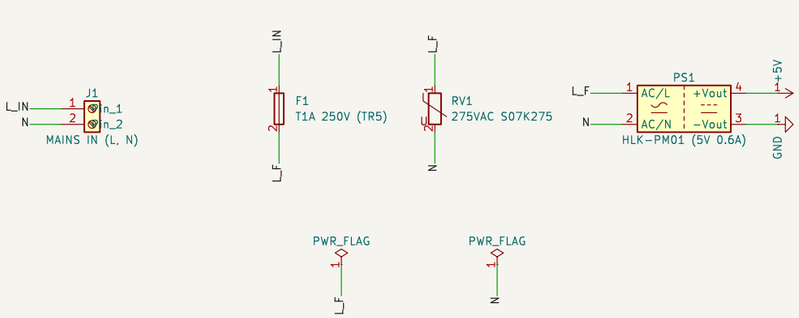
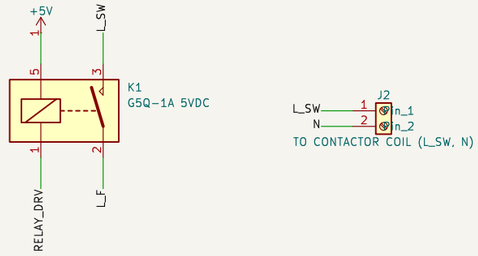
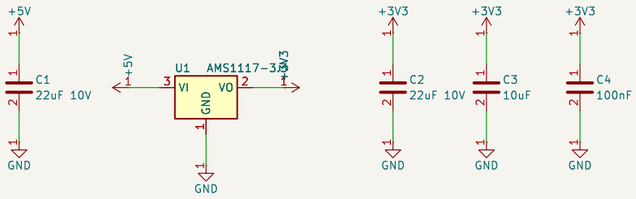
MAINS 230 VAC — DANGER



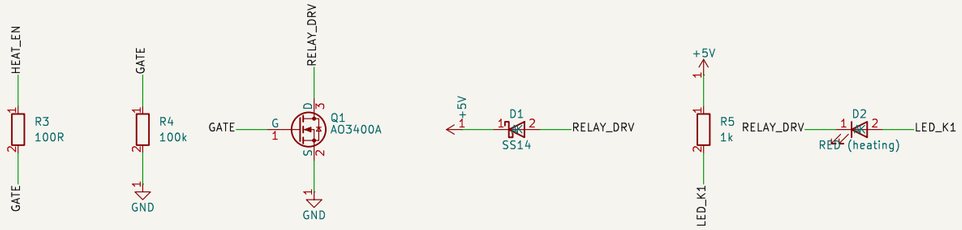


K1 contact switches only the contactor coil (A1/A2).
Heater load goes through the contactor, never through K1.

POWER 5 V -> 3.3 V

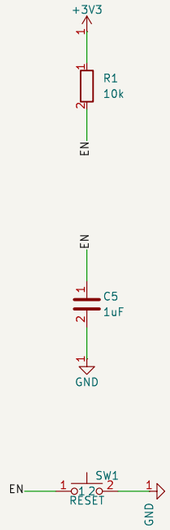


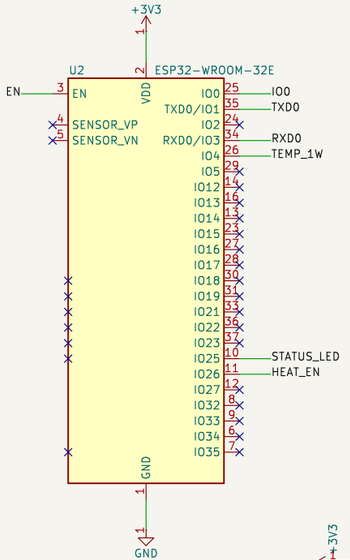
RELAY DRIVER

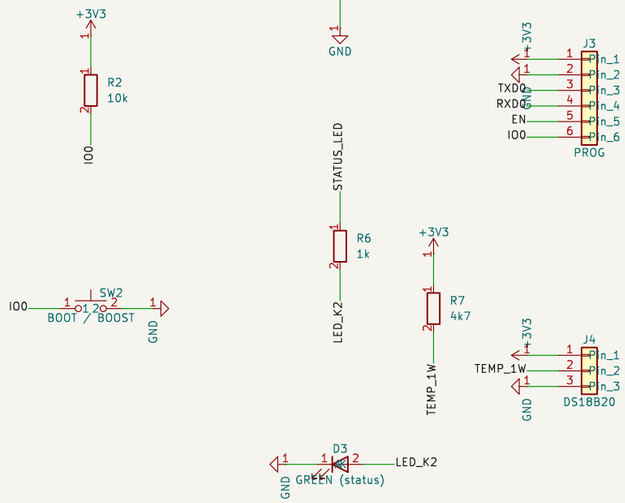


R4 holds the relay OFF while the ESP32 boots or resets.

ESP32 + UI







J3: connect a 3.3 V USB−UART adapter with MAINS DISCONNECTED.
Hold BOOT, tap RESET to flash. Later updates over WiFi (OTA).
SW2 doubles as a manual 'boost' button after boot.
J3 pins: 1=3V3 2=GND 3=TXD0 4=RXD0 5=EN 6=IO0. J4 (optional): 1=3V3 2=DATA 3=GND.

Mains wiring and contactor installation by a qualified electrician.
Switches an external contactor coil (230 VAC). Tank thermostat/limiter stays in series.
ESP32−WROOM−32E, WiFi, Nord Pool spot price

Sheet: /
File: spot_heater.kicad_sch

Title: Spot−price water heater controller

Size: A3	Date: 2026−10−20	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 1/1